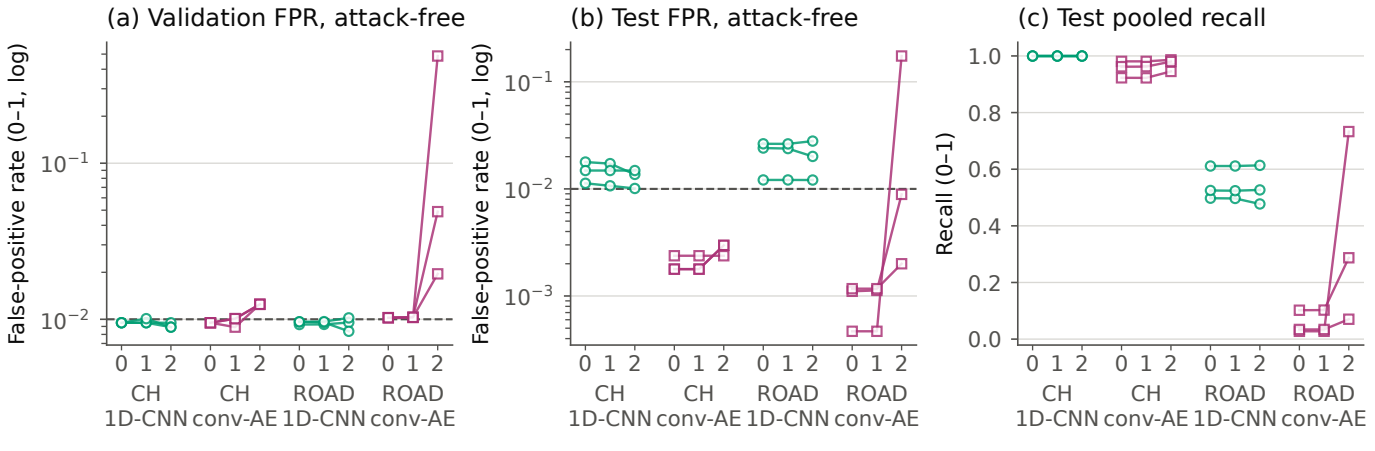
(a) Validation FPR, attack-free
(b) Test FPR, attack-free
(c) Test pooled recall
False-positive rate (0–1, log)
Recall (0–1)
10^{-1}
10^{-2}
10^{-3}
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2
CH 1D-CNN
CH conv-AE
ROAD 1D-CNN
ROAD conv-AE
x: quantization level k (0 = fp32, 1 = fp16, 2 = INT8); Q0 threshold kept for every k
1D-CNN: one line per seed
conv-AE: one line per seed
target FPR of the threshold (1%, val)